

Recommendation System

Technical Report

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Live Demo: huggingface.co/spaces/deepanvishal/serko-recsys

GitHub: github.com/deepanvishal/RecSys

W&B: wandb.ai/deepanvishal-cvs-health/serko-recsys-movielens

1. The Product

The system is a fully functional recommendation engine with a browser-based UI. It runs five recommendation models, routes each user to the right model based on their watch history, and surfaces results through an interface designed to show how recommendation logic changes depending on what we know about the user.

The UI has four tabs:

- **For You — Main Demo:** A Netflix-style personalised homepage. This is the primary demonstration of the full system. Described in detail below sections
- **Recommendations:** Select a movie from a dropdown. See what all five models recommend next. Optionally filter Two-Tower output by user demographics. Shows the selected movie's interaction distribution by age and gender.
- **Model Bias:** Popularity ratio, demographic hit rate gaps, and genre concentration across models.
- **Cold Start:** Two scenarios — a new movie with no ratings yet, and a new user with no watch history. Shows how the system handles both using content signals and demographic trends.

2. For You — The Main Demo

The For You tab shows what the system would look like in production. It is a single-page personalised homepage that renders five recommendation rows for a selected user profile. The rows and the models powering them change depending on how much interaction history the user has.

Select a user profile from the dropdown — five profiles are available, ranging from a power user with 200+ watched movies to a brand-new user with no history. The page rebuilds immediately to reflect that user's data.

How the rows change based on available data

Row	Label shown	User with history (>5 movies)	New user (0 movies)
1	Based on your watch history / Trending	SASRec — sequential model predicting next watch from full history	Trending — globally most-watched movies (no personal signal available)
2	Trending / Popular in [Genre]	Trending — globally most popular items	Popular in [top genre for this age+gender] — most watched genre among users with same demographics
3	Because you watched [Genre 1] / Popular in [Genre]	Two-Tower top 200 → filtered to user's most-watched genre → top 5	Popular in [2nd genre for this age+gender]
4	Because you watched [Genre 2] / Popular in [Genre]	Two-Tower top 200 → filtered to user's 2nd most-watched genre → top 5	Popular in [top genre for this gender] — different from rows 2 and 3
5	New arrivals	Simulated — 10 real catalog movies treated as recently added (fixed seed)	Same

Each movie card shows title, genres, and year. Rows are displayed horizontally in a Netflix-style layout.

The five user profiles in the dropdown:

Profile	Type	History	What changes on the page
Alex — Power User	Warm	200+ movies	Row 1 is SASRec. Genre rows reflect deep, established preferences.
Sarah — Movie Enthusiast	Warm	40–60 movies	Row 1 is SASRec. Genre rows reflect Drama/Romance preferences.
Marcus — Casual Viewer	Warm	15–25 movies	Row 1 is SASRec with shorter history.
Priya — Light Viewer	Sparse	4 movies	Row 1 is absent (not enough history for SASRec). Trending leads.
New User	Cold	0 movies	Gender + age selector appears. All rows are demographic-driven.

3. Dataset

The system is built on MovieLens 1M, a standard recommendation benchmark from GroupLens Research.

Stat	Value
Total ratings	1,000,209
Users	6,040

Stat	Value
Movies	3,706
Average ratings/user	165
Average ratings/movie	270
Rating scale	1–5 stars (binarised to interactions)
User attributes	Age, gender, occupation
Item attributes	Title, genres (18 categories), year

Evaluation: leave-one-out split. The last interaction per user is held out as the test item. Models are evaluated by ranking the test item against all 3,706 movies in the catalog — this is a stricter protocol than the sampled evaluation (99 negatives) used in many published papers. Our numbers are directly comparable to or better than published results under the stricter protocol.

4. Models

Collaborative Filtering

Item-item cosine similarity using sparse interaction matrices. Simple and interpretable. $HR@10 = 0.083$. User-user CF marginally outperforms at the same metric, consistent with the dataset's dense histories where user-space similarity carries stronger signal than item co-occurrence.

SVD / Matrix Factorisation

Alternating Least Squares with 128 latent factors. The key production feature is fold-in: a new or sparse user's partial history is used to compute a user embedding on the fly in under 100ms, without any model retraining. $HR@10 = 0.102$.

Two-Tower

Dual encoder: a user tower (user ID embedding + GRU over watch history + demographics) and an item tower (item ID embedding + sentence transformer on title+genres + genre vector). Both towers produce 128-dim L2-normalised embeddings compared via dot product.

$HR@10 = 0.065$ on this dataset — below CF. On dense data, collaborative signal is stronger than content signal. Two-Tower's value is in cold-item scenarios: a new movie can be embedded from its title and genres immediately, with no retraining needed.

SASRec

Self-attentive sequential recommendation. Treats each user's watch history as a sequence and uses transformer self-attention to predict what they will watch next. $HR@10 = 0.333$, $NDCG@10 = 0.187$. This is the primary model for users with sufficient history — 3.26× better than SVD because it captures how preferences shift over time.

BERT4Rec

Bidirectional transformer with masked item prediction. Same architecture size as SASRec for fair comparison. HR@10 = 0.042. Early stopping fired before convergence — published BERT4Rec results on this dataset require 200+ training epochs. Included as an experimental comparison model.

5. Model Results

Model	HR@10	NDCG@10	Role
SASRec	0.3328	0.1873	Primary — warm users (>5 interactions)
SVD (f=128)	0.1020	0.0487	Tier 2 — sparse users (1–5 interactions)
UserUserCF	0.0830	0.0404	Tier 2 — fallback
ItemItemCF	0.0816	0.0390	Item-item similarity
Two-Tower	0.0647	0.0315	Cold-item retrieval
BERT4Rec	0.0417	0.0199	Experimental

6. Tiered Inference Engine

Every request is routed to the most appropriate model based on how much we know about the user.

Tier	Trigger	Model	Latency
1	0 interactions	Popularity + reranker	<10ms
2	1–5 interactions	CF + SVD + reranker	~50ms
3	>5 interactions	SASRec (no reranker)	~2ms (GPU)
Cold item	New item, no ratings	Two-Tower content encoding	~70ms

7. LightGBM Reranker

For Tier 1 and Tier 2, a LightGBM binary classifier reranks candidates before serving. It is a second-stage model that takes the top-100 candidates from the retrieval step and scores them more carefully.

Input data

One row per (user, candidate item) pair. Features:

- **retrieval_score**: raw score from the upstream retrieval model (CF similarity or SVD dot product)
- **item_popularity**: how many times this item was interacted with in training data

- **genre_affinity**: dot product between the item's genre vector and the user's historical genre profile
- **gender_enc, age_enc, occupation**: user demographic features
- **n_history**: number of items in the user's training history
- **user_genre_[18 genres]**: fraction of the user's history belonging to each genre

Target data

Binary label: 1 if the candidate item is the held-out ground truth item for that user, 0 for all other candidates. Training data was built by running CF, SVD, and Two-Tower retrieval over all validation users and labelling the held-out item as positive.

Validation AUC: 0.7417. Top features by importance: item_popularity (1193), genre_affinity (957), retrieval_score (744). The reranker is not applied to SASRec or BERT4Rec — sequential models encode preference patterns that a popularity-biased reranker trained on shallow model candidates would degrade.

8. Cold Start

New item

When a new movie arrives with no ratings, the Two-Tower item tower encodes it from title (sentence transformer) and genres (18-dim multi-hot). The 128-dim embedding is immediately available for similarity search against all user embeddings. No retraining required.

New user

Zero history → Tier 1 (popularity). When demographics are available, the system shows trending items among users with the same demographic profile, and Two-Tower recommendations using demographics alone (empty history, age and gender as the only signal).

9. Bias Audit

Metric	Value	Interpretation
SASRec popularity ratio	3.18×	Recommends items 3× more popular than catalog average
SVD popularity ratio	4.24×	SVD is more popularity-biased than SASRec
Gender gap (M-F HR@10)	+0.025	Male 34.0% vs Female 31.5% — data volume asymmetry
Age HR@10 range	0.110	Under-18: 25.7% (worst), 50–55: 36.7% (best)
Top genre in recs	Comedy 35.6%	Mirrors catalog distribution — not pathological

SASRec's popularity ratio (3.18×) is lower than SVD's (4.24×) — the more expressive model is also the less popularity-biased one. The under-18 cohort underperforms by 7–11 percentage points — a small training population (222 users, 3.7% of total), not a model failure, but a real deployment consideration.

10. Technical Stack

Component	Technology
Models	PyTorch, implicit (ALS/CF), LightGBM, sentence-transformers
Vector search	FAISS (FlatIP, 128-dim embeddings)
API	FastAPI + uvicorn, 6 endpoints
UI	Streamlit, 4 tabs
Experiment tracking	Weights & Biases
Data format	Parquet

11. Links

- **Live demo:** huggingface.co/spaces/deepanvishal/serko-recsys
- **GitHub:** github.com/deepanvishal/RecSys
- **W&B training runs:** wandb.ai/deepanvishal-cvs-health/serko-recsys-movielens